

Allana Tavares Bastos

◆ allanatavares@dcc.ufmg.br ◆ +55 (31) 99538-1108 ◆ Personal webpage: allana-tb.github.io ◆

Education

PhD in Computer Science 03/2025 - 03/2029

Federal University of Minas Gerais (UFMG)

Advisor: Professor Martín G. Ravetti

Recipient of CAPES PhD Scholarship (2025 - 2029)

MsC in Computer Science 08/2022 - 12/2024

Federal University of Minas Gerais (UFMG)

GPA: 4.6/5.0

Advisor: Professor Martín G. Ravetti

Recipient of CAPES Masters Scholarship (2022 - 2024)

Dissertation: Diversity-Aware Graphormer: Insights from Hate Speech Detection on 4chan

Graduate Diploma in Systems Analysis and Development 06/2021 - 12/2023

Pontifical Catholic University of Minas Gerais (PUC Minas)

GPA: 4.9/5.0

Awarded Best Project of a Mobile App in class (12/2022)

Bachelor of Science in Business Administration 06/2017 - 09/2021

Federal University of Minas Gerais (UFMG)

GPA: 4.6/5.0

Final paper: Robustness of the public transport network of Belo Horizonte, advised by Professor Tiago A. Schieber

Relevant subjects: Software Engineering, Artificial Intelligence Ethics, Stochastic Programming, Design and Analysis of Algorithms, Heuristics and Metaheuristics, Geoprocessing, Machine Learning

Skills

Programming: Python, R, SQL, Html, CSS and JavaScript, Git, C++, Kubernetes

Languages: Portuguese (fluent), English (fluent), Spanish (intermediary), and French (elementary)

Work Experience

Research Staff Member and Developer 04/2023 - 03/2025

Future Lab

- Elicited requirements and developed the backend and API of Flautim, a cloud machine learning platform for Federated Learning
- Tested the platform with federated and unfederated machine learning projects

Data Science Internship 04/2022 - 08/2022

RaiaDrogasil (RD)

- Developed an implicit recommendation system for product pages, based on product's category
- Optimized NLP-based categorization of products' ratings and comments from 60% to 85% accuracy, providing better insights for the customers relations teams and raising internal performance indicators
- Tech Stack: Python | Kedro | SQL | AWS | GitLab

Special Projects Internship 09/2019 - 07/2021

Municipality of Belo Horizonte

- Cleaned up personnel information in proprietary software, and emitted, checked and altered data in SQL reports

Research Experience

Research Fellowship 03/2020 - 08/2021

Administrative Sciences Department - UFMG | Advisor: Professor Tiago A. Schieber

- Developed a research in Network Science on the detection of anomalies in bidding processes, constructing a measure to check abnormal interactions in the networks, using R
- Took part in a research group on modeling the pandemic with 91% accuracy, using Artificial Intelligence, Machine Learning techniques and Network Science concepts

Research Fellowship**08/2017 - 08/2019***PET Administration - UFMG | Advisor: Professor Deise L. Ferraz*

- Collaborated in a self-organized group with demands in research and extension, including the planning and execution of workshops, talks, radio programs and qualitative research
- Was the editor in chief of a student academic magazine, Multifaces, from 01/2018 to 08/2019, leading an editorial team with 7 peers in the publishing process of 2 editions

Junior Research Fellowship**07/2015 - 12/2016***CEFET-MG | Advisor: Christiano Rocha, MsC*

- Was oriented in a research in enumerability and infinity, which resulted in a presentation in a local conference organized by CEFET-MG